Exp 10

Aim: Explore the Cryptocurrency Landscape

Theory:

Cryptocurrency Landscape – Theory
:
Cryptocurrencies are decentralized digital assets that operate on blockchain technology, enabling secure, transparent, and immutable transactions without the need for a central authority. These systems rely on distributed ledgers where transactions are validated through consensus mechanisms and recorded across a network of nodes.

Data Sources
Reliable and diverse data sources are essential for understanding cryptocurrency markets:
- Financial Data Platforms – Services like CoinGecko and CoinMarketCap provide structured data such as market capitalization, circulating supply, trading volume, and historical price trends.
- News Aggregators – Platforms such as CryptoPanic collect real-time news from multiple sources, helping track major events impacting the market.
- Social Sentiment Analytics – Tools like LunarCrush analyze social media activity to measure public sentiment and engagement.
- Regulatory Sources – Official portals like the U.S. Securities and Exchange Commission and frameworks such as MiCA provide insights into legal policies and compliance requirements.

Types of Cryptocurrencies
The cryptocurrency ecosystem consists of various categories based on functionality: ● Store of Value (SoV) – Bitcoin (BTC), known for limited supply and long-term value preservation.
- Utility Tokens – Ether (ETH) and Chainlink (LINK), used for network operations and smart contract execution.
- Stablecoins – Tether (USDT) and USD Coin (USDC), designed to maintain price stability by being pegged to fiat currencies.
- Governance Tokens – Uniswap (UNI) and Aave (AAVE), enabling users to participate in decentralized decision-making.

Key Characteristics

Cryptocurrencies exhibit the following features:

- Decentralization – No central authority controls the system.
- Transparency – Transactions are publicly recorded on the blockchain.
- Immutability – Data cannot be altered once confirmed.
- Security – Cryptographic algorithms ensure data integrity and safety.

Market Dynamics

The cryptocurrency market is influenced by multiple factors:

- Volatility – Prices fluctuate significantly due to demand, sentiment, and external events.
- Liquidity – Depends on trading volume and market participation.
- Market Sentiment – News, social media, and investor behavior impact price trends.
- Regulation – Government policies and frameworks influence adoption and stability.

Blockchain Technology Advancements

Modern blockchain systems are evolving to overcome limitations:

- Consensus Mechanisms – Shift from Proof of Work (PoW) to Proof of Stake (PoS) for efficiency.
- Scalability Solutions – Layer-2 technologies and sharding improve transaction speed.
- Interoperability – Enables communication between different blockchain networks. ● Privacy Enhancements – Zero-Knowledge Proofs (ZKP) allow secure verification without revealing sensitive data.

Adoption and Emerging Trends

Cryptocurrency adoption continues to grow across sectors:

- Digital Payments – Companies like Stripe and Shopify support crypto transactions.
- Decentralized Finance (DeFi) – Financial services operate without intermediaries.
- Non-Custodial Wallets – MetaMask and Phantom give users full asset control. ● Asset Tokenization – Real-world assets are increasingly represented on blockchain networks.

Utilizing Api for analysis and visualization :

Code:

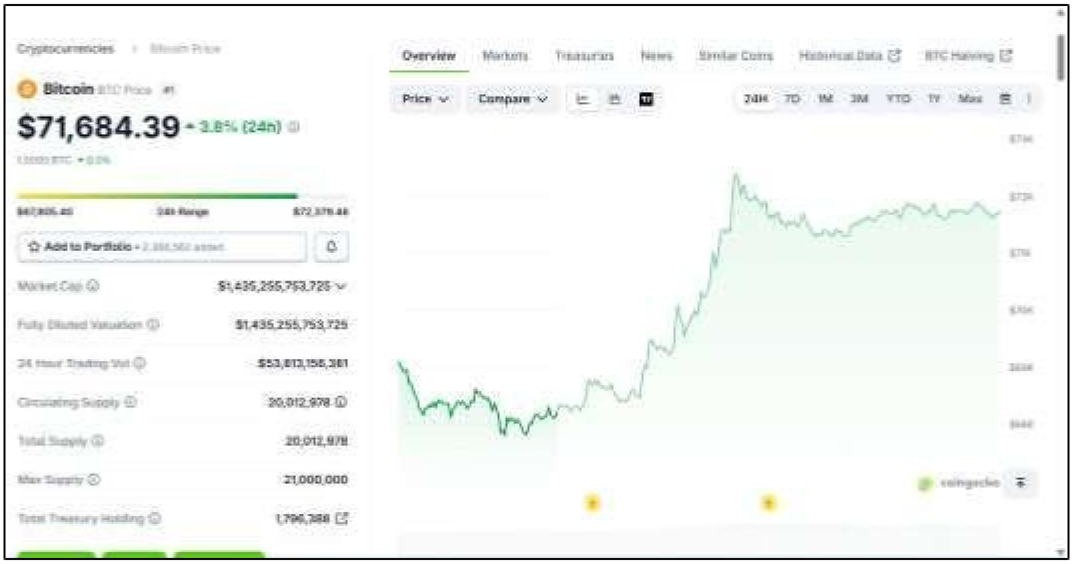
```
!pip install requests pandas matplotlib

# Imports import
requests import pandas
as pd import
matplotlib.pyplot as plt

# ----- CONFIG -----
API_KEY = "api-key"
COIN = "BTC"
CURRENCY = "USD"
DAYS = 180

# -------- FETCH FUNCTION       def
fetch_data(coin, currency, days):
  url = "https://min-api.cryptocompare.com/data/v2/histoday"

  params = {
    "fsym":
    coin,
    "tsym": currency,
    "limit": days,
    "api_key":
    API_KEY
  }

  res = requests.get(url,
params=params) data = res.json()

  if data["Response"] != "Success":
print("Error:", data.get("Message"))
return None

  return data["Data"]["Data"]

# -------- LOAD DATA       raw =
fetch_data(COIN, CURRENCY, DAYS) df =
pd.DataFrame(raw)
```

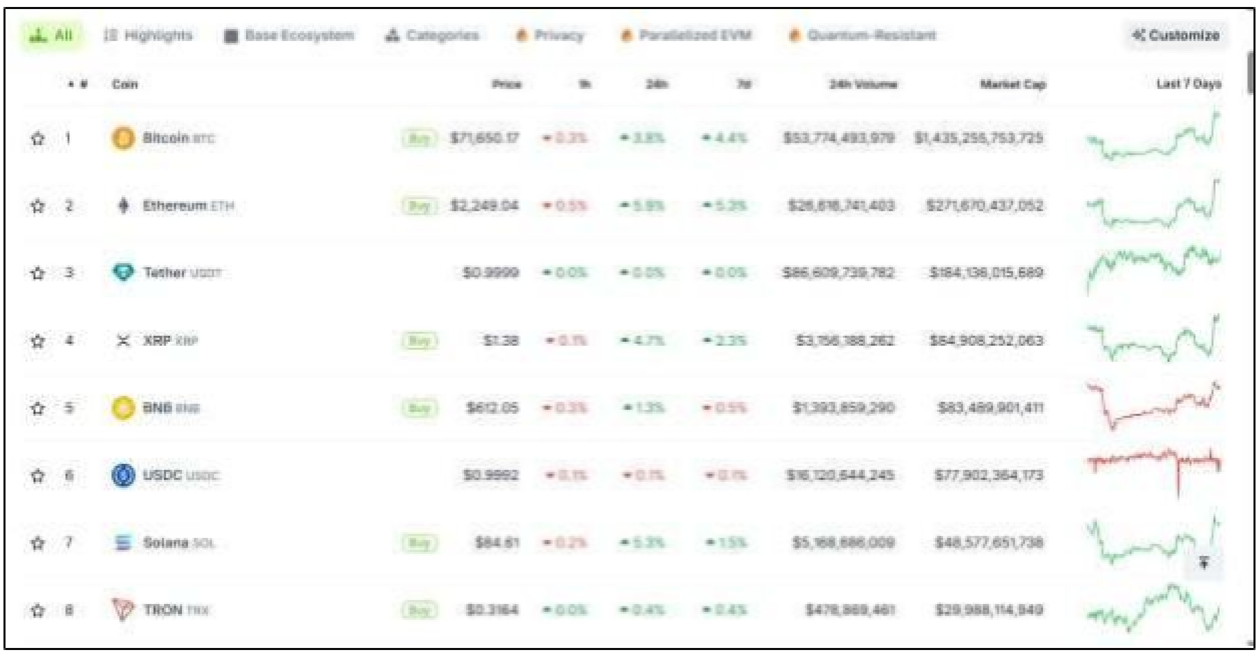
```
# Time formatting
df["date"] = pd.to_datetime(df["time"], unit="s")
df.set_index("date", inplace=True)

# Core columns df["price"]
= df["close"] df["volume"] =
df["volumeto"]

# -------- CALCULATIONS        df["returns"] =
df["price"].pct_change() df["MA_7"] =
df["price"].rolling(7).mean() df["MA_21"] =
df["price"].rolling(21).mean()
df["cumulative_return"]
= (1 + df["returns"]).cumprod() df["volatility"] =
df["returns"].rolling(7).std() * 100

# -------- PLOTTING (6 GRAPHS)      plt.figure(figsize=(16,
12))

# 1. Price Trend
plt.subplot(3, 2, 1)
plt.plot(df["price"])
plt.title("1. Price Trend")
plt.grid()

# 2. Price vs Volume plt.subplot(3, 2, 2)
plt.plot(df["price"], label="Price") plt.bar(df.index,
df["volume"], alpha=0.3, label="Volume")
plt.legend() plt.title("2. Price vs Volume")

# 3. Returns Distribution
plt.subplot(3, 2, 3)
df["returns"].dropna().hist(bins=40)
plt.title("3. Returns Distribution")

# 4. Moving Averages
plt.subplot(3, 2, 4)
```

```
plt.plot(df["price"], label="Price", alpha=0.5)
plt.plot(df["MA_7"], label="7-Day MA")
plt.plot(df["MA_21"], label="21-Day MA") plt.legend()
plt.title("4. Moving Averages")

# 5. Cumulative Growth
plt.subplot(3, 2, 5)
plt.plot(df["cumulative_return"])
plt.title("5. Cumulative Returns")

# 6. Volatility
plt.subplot(3, 2, 6)
plt.plot(df["volatility"])
plt.title("6. Volatility
(%)")

plt.tight_layout() plt.show()
```

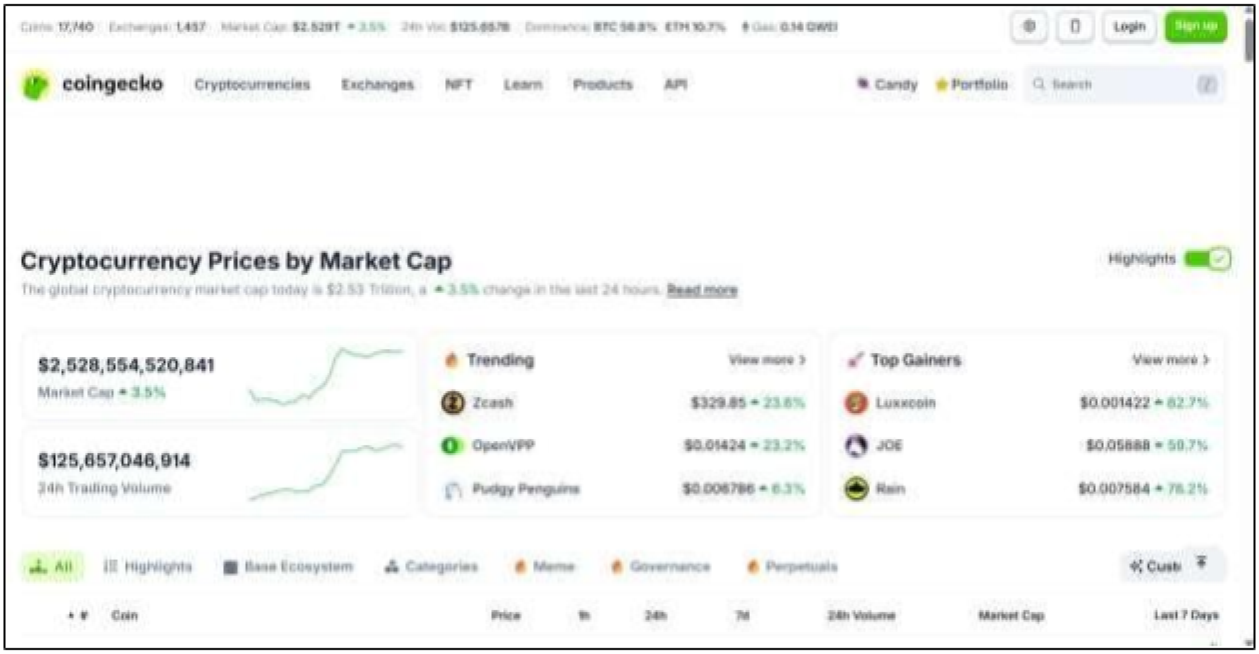
D20A 59 Jai Talreja Blockchain Lab

Output:-

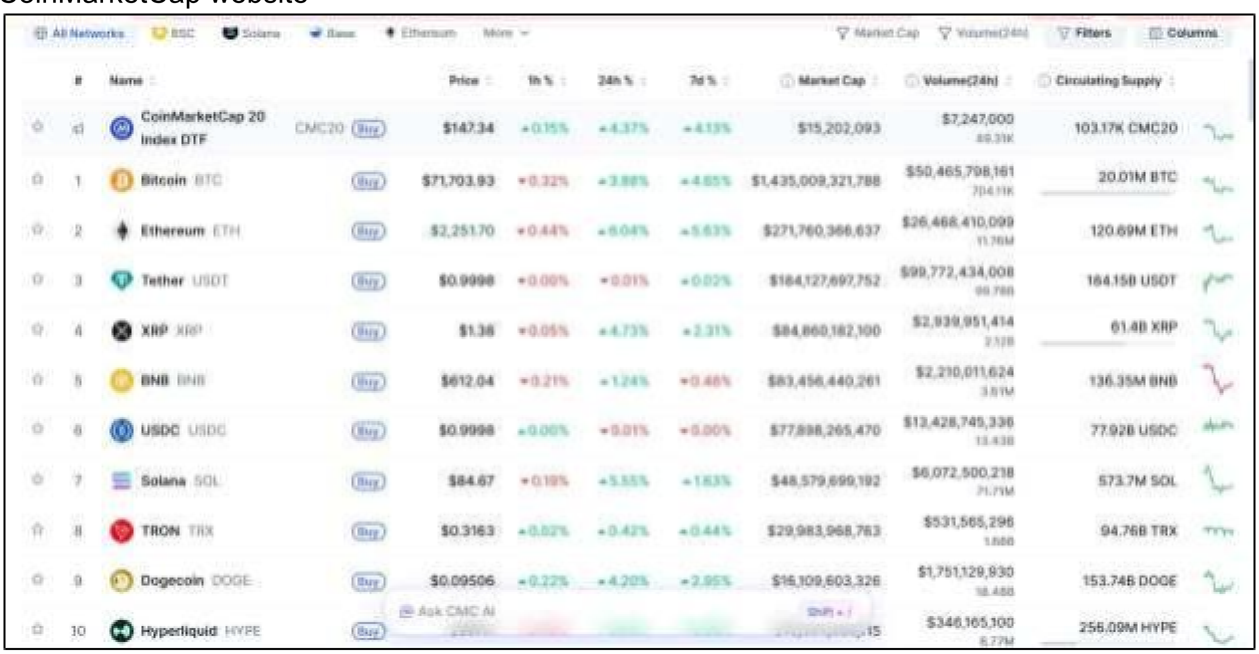




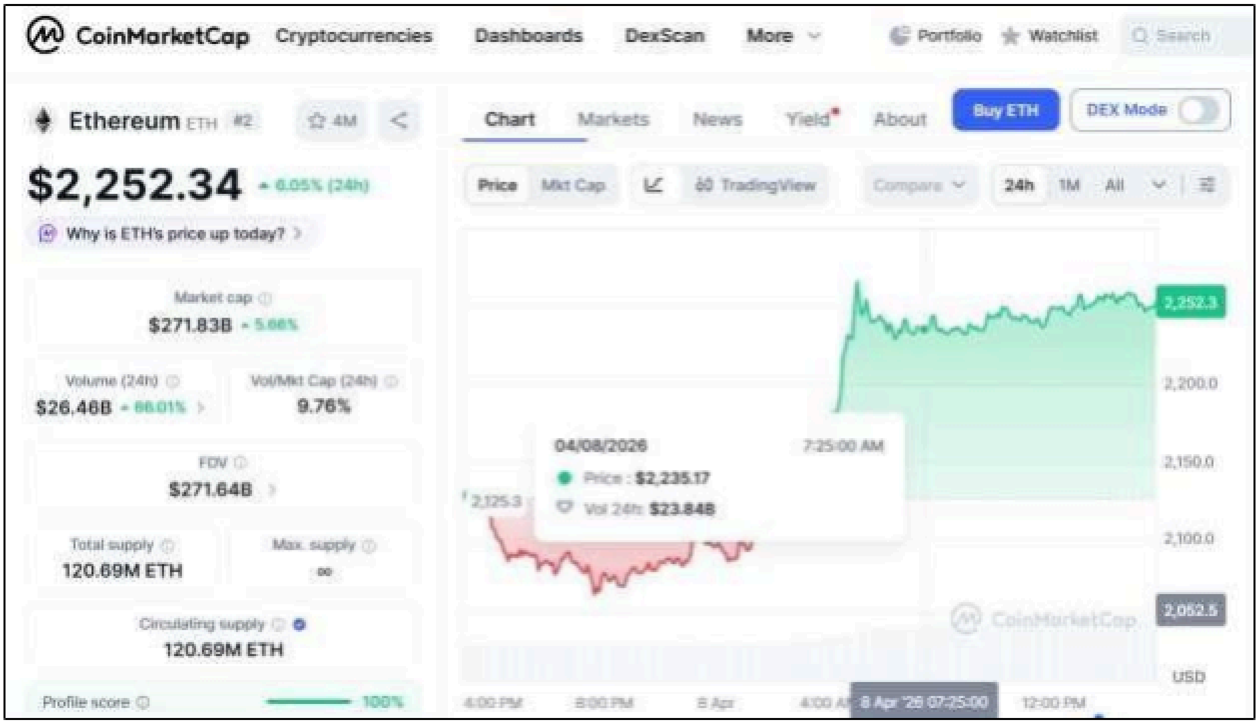
D20A 59 Jai Talreja Blockchain Lab



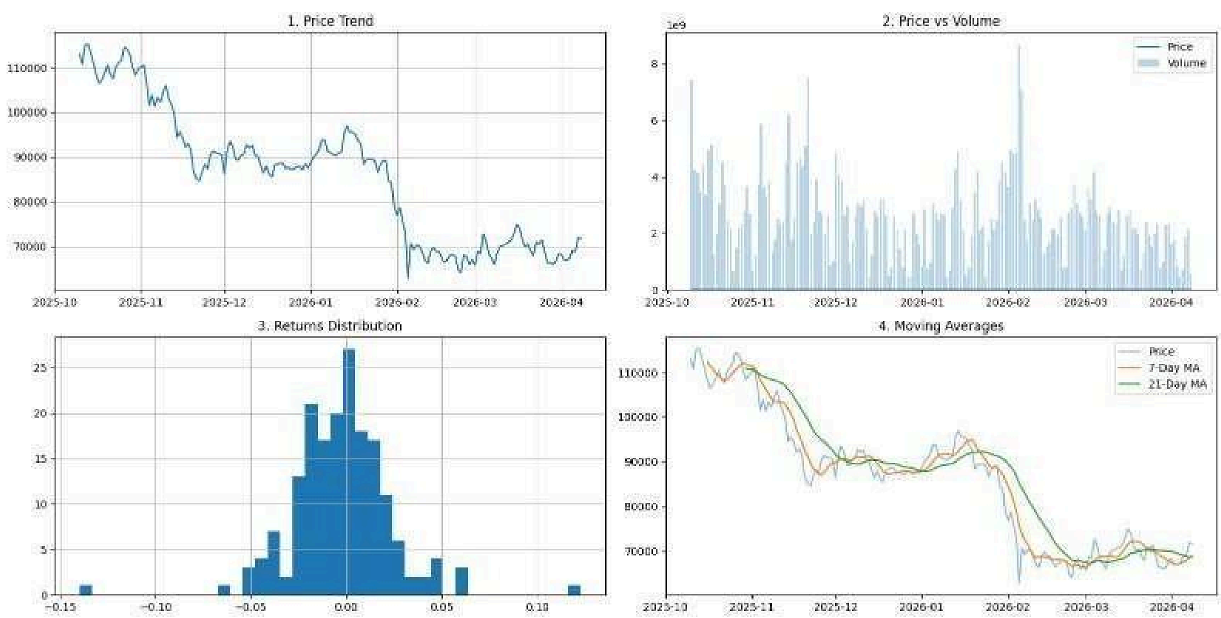
CoinMarketCap website

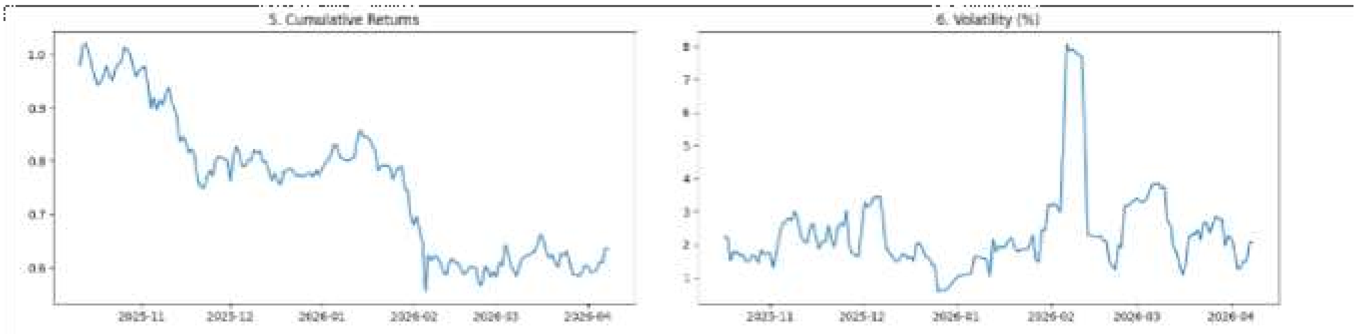


D20A 59 Jai Talreja Blockchain Lab



Dashboard :





Conclusion:-

By integrating automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, a comprehensive view of the cryptocurrency ecosystem can be achieved. This approach highlights the relationship between blockchain's underlying technical foundations and real-world market behavior, offering valuable insights into current trends as well as potential future developments in the digital asset industry.